Chemistry · Unit 1C · Level 2

Conductivity Lab: What Lets Electricity Through?



TODAY'S PLAN

- 👥 Jobs
- ⚠️ Safety
- 🌱 Steps
- ✅ Check
- 📝 Reflect

Get ready

Jobs



Tester

Touches the two tester ends to the material (both ends in the cup for liquids).



Bulb watcher

Says: bright, dim, or off.



Recorder

Writes the result in the table.



Dryer

Dries the tester ends with a paper towel between tests.

Safety

Be Safe



🥽 Goggles on. Copper sulfate and salt solutions can sting eyes.

🚫 Never taste anything in a lab, even salt or sugar water.

🔌 Only use the low-voltage tester. Never put anything in a wall outlet.

🧼 Wash hands after.

Steps

Steps 1 of 2



1. 🥽 Goggles on. Check the tester lights when its two ends touch.
2. 🔮 **Predict**: will each thing conduct? Write yes or no.
3. 🔩 **Test the solids**: foil/copper, wax, sugar, salt crystals.
4. 💧 **Test the liquids**: plain water, then salt water (1%, 5%, 10% if available).

Steps

Steps 2 of 2



1. 🍬 **Test** sugar water.
2. 💡 **Record**: bright, dim, or off.
3. 🧻 Dry the tester between every test.
4. 📝 **Sort**: conductors vs. insulators. What do the conductors have in common?

🃏 Check 1 of 2



More salt in the water makes the bulb…

Hold up 1, 2, or 3:

1️⃣ Brighter

2️⃣ Dimmer

3️⃣ Turn off

✅ Answer · Card 1 of 2



Brighter

More ions, more current.

🃏 Check 2 of 2



Sugar water does not conduct because sugar…

Hold up 1, 2, or 3:

1️⃣ Is too sweet

2️⃣ Is a metal

3️⃣ Does not make ions

✅ Answer · Card 2 of 2



Does not make ions

Sugar dissolves but stays neutral.

Wrap-Up

Clean Up + Reflect



- 🥽 Goggles back
- 🧼 Wash hands
- 📝 Finish the table and CER